

Additive models

Data Mining - CdL CLAMSES

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The Great Barrier Reef

- In this unit we will cover the following **topics**:
 - Generalized additive models (GAMs)
- We have seen that **fully nonparametric** methods are plagued by the **curse of dimensionality**.
- GAMs are **semi-parametric** approaches that keep the model complexity under control so that:
 - they are more flexible than linear models;
 - they are not hugely impacted by the curse of dimensionality.
- The running example is about **trawl data** from the **Great Barrier Reef**.

An ecological application

The **trawl** dataset

- We consider the **trawl** dataset, which refers to a **survey** of the **fauna** on the sea bed lying between the coast of northern Queensland and the **Great Barrier Reef**.
- The **response** variable is **Score**, which is a standardized numeric quantity measuring the amount of fishes caught on a given location.
- We want to **predict** the **catch score**, as a function of a few covariates:
 - the **Latitude** and **Longitude** of the sampling position. The longitude can be seen as a proxy of the distance from the coast in this specific experiment;
 - the **Depth** of the sea on the sampling position;
 - the **Zone** of the sampling region, either open or closed to **commercial fishing**;
 - the **Year** of the sampling, which can be either **1992** or **1993**.
- Having remove a few observations due to missingness, we split the data into **training** (119 obs.) and **test** set (30 obs.). The full **trawl** dataset is available in the **sm** R package.

The trawl dataset

Getting started: linear models

- Let us begin our analysis by trying to predict the **Score** using a **linear model** of the form

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}, \quad i = 1, \dots, n,$$

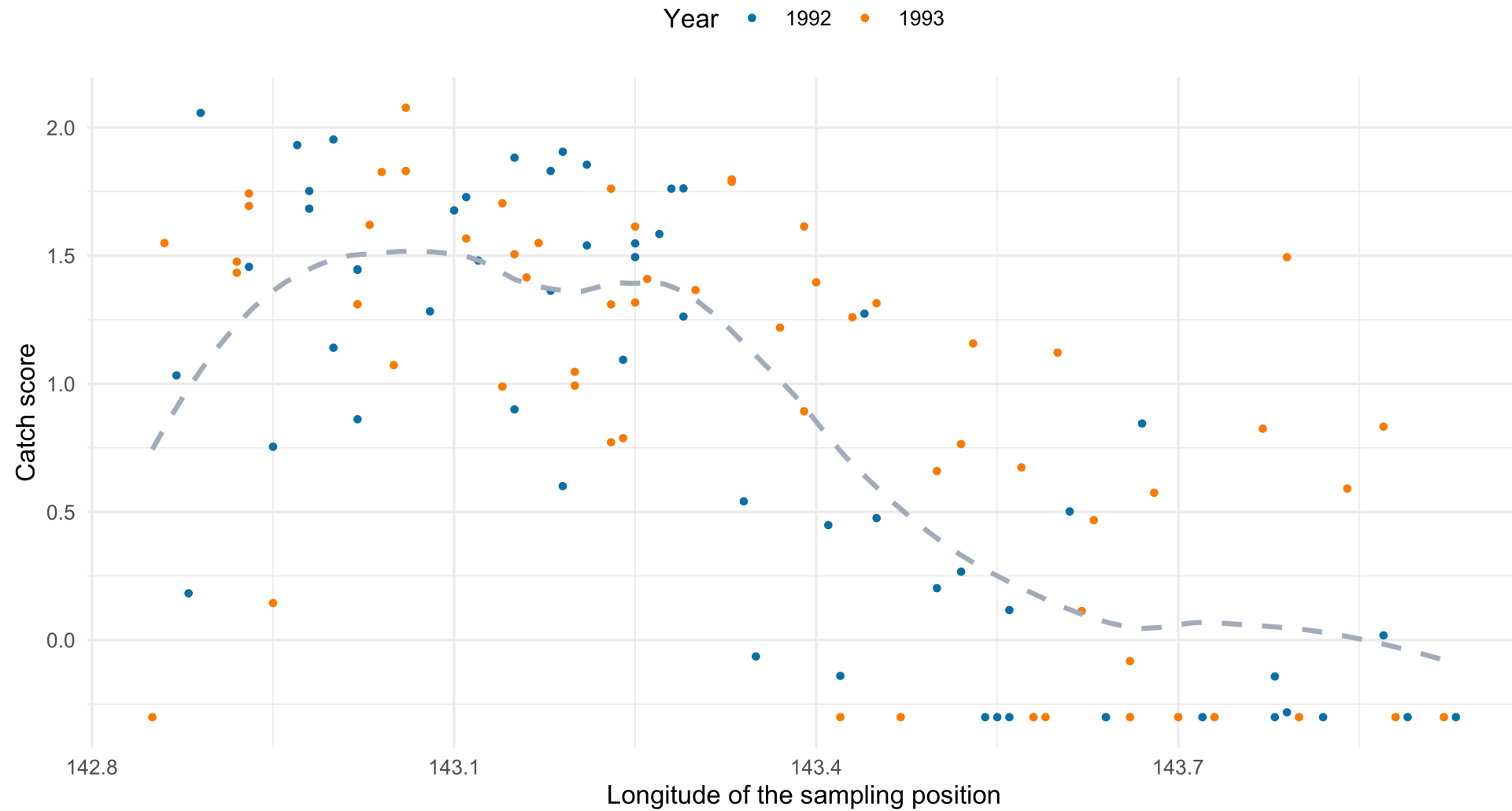
- The above values correspond to the variables of the **trawl** dataset, so that

$$\begin{aligned} \text{Score}_i = & \beta_0 + \beta_1 \text{Latitude}_i + \beta_2 \text{Longitude}_i + \\ & + \beta_3 \text{Depth}_i + \beta_4 I(\text{Zone}_i = \text{Closed}) + \beta_5 I(\text{Year}_i = 1993). \end{aligned}$$

- Such a model can be estimated using **ordinary least squares**, resulting in:

term	estimate	std.error	statistic	p.value
(Intercept)	297.690	26.821	11.099	0.000
Latitude	0.256	0.222	1.151	0.252
Longitude	-2.054	0.187	-10.955	0.000
Depth	0.020	0.007	3.003	0.003
Zone_Closed	-0.116	0.102	-1.143	0.255
Year_1993	0.127	0.103	1.242	0.217

Scatterplot with **loess** estimate



Comments and criticism of linear models

- Is this a good model?
- Granted that every model is just an approximation of reality, it is undeniable that there are some **problematic** aspects.
- By simple graphical inspection, it seems that the relationship between **Score** and **Longitude** is **non-linear**.
- Also, an **interaction effect** between **Year** and **Longitude** could be present.
- These considerations support the idea that a **nonparametric approach** might be more appropriate.
- However, the number of covariates is $p = 5$ and therefore a fully nonparametric estimation would **not** be **feasible**, because of the curse of dimensionality.
- We need a simplified modelling strategy, that accounts for non-linearities but at the same time is not fully flexible.

Generalized additive models (GAM)

The ANOVA decomposition of a function

- We seek for an estimate of (a suitable transformation of) the **mean function**, namely

$$g^{-1}\{\mathbb{E}(Y_i)\} = f(x_{i1}, \dots, x_{ip}),$$

where $g^{-1}(\cdot)$ is the so-called **link function**.

- The **unknown** multivariate function $f(\mathbf{x}) = f(x_1, \dots, x_p) : \mathbb{R}^p \rightarrow \mathbb{R}$ is **too complex**. However, the following **decomposition** holds

$$f(\mathbf{x}) = \beta_0 + \underbrace{\sum_{j=1}^p f_j(x_j)}_{\text{Main effect}} + \underbrace{\sum_{j=1}^p \sum_{k < j} f_{jk}(x_j, x_k)}_{\text{Interaction effect}} + \underbrace{\sum_{j=1}^p \sum_{k < j} \sum_{h < k < j} f_{jkh}(x_j, x_k, x_h) + \dots}_{\text{Higher order interaction}}.$$

- By imposing **suitable constraints**, this decomposition can be made **unique**.
- More importantly, this decomposition gives us an intuition on how to build non-linear models with a simplified structure.

Generalized additive models (GAM)

- A **generalized additive model** (GAM) presumes a representation of the following type:

$$f(\mathbf{x}_i) = \beta_0 + f_1(x_{i1}) + \cdots + f_p(x_{ip}) = \beta_0 + \sum_{j=1}^p f_j(x_{ij}), \quad i = 1, \dots, n,$$

where $f_1, \dots, f_p$ are **smooth univariate** functions with a potentially non-linear behavior.

- In GAMs we include only the **main effects** and we **exclude** the **interactions** terms.
- Generalized linear models (GLMs) are a **special case** of GAMs, in which $f_j(x_{ij}) = \beta_j x_{ij}$.
- To avoid what is essentially a problem of **model identifiability**, it is necessary for the various f_j to be centered around 0, that is

$$\sum_{i=1}^n f_j(x_{ij}) = 0, \quad j = 1, \dots, p.$$

The backfitting algorithm I

- There exist several strategies for **estimating** the **unknown functions** $f_1, \dots, f_p$. One of them, called **backfitting**, is particularly appealing because of its elegance and generality.
- Suppose we model each $f_j(x) = \sum_{m=1}^{M_j} \beta_{mj} h_{mj}(x)$ with a **basis expansion**, for example using **regression splines**.
- In a **regression problem** we need to minimize, over the unknown β parameters, the loss

$$\sum_{i=1}^n \left\{ y_i - \beta_0 - \sum_{j=1}^p f_j(x_{ij}) \right\}^2$$

subject to the constraint $\sum_{i=1}^n f_j(x_{ij}) = 0$.

- When f_j are regression splines, the above loss can be **minimized** using **least squares**. The identifiability issue could be handled by removing the intercept term from each spline basis.
- However, here we consider an **alternative** and **iterative** minimization method, which is similar to the coordinate descent algorithm we employed for the elastic-net.

The backfitting algorithm II

- Now, let us re-arrange the term in the squared loss as follows:

$$\sum_{i=1}^n \left\{ y_i - \beta_0 - \sum_{k \neq j} f_k(x_{ik}) - f_j(x_{ij}) \right\}^2,$$

where the highlighted terms are sometimes called **partial residuals**.

- Hence, we can repeatedly and iteratively fit a **univariate smoothing** model for f_j using the **partial residuals** as **response**, keeping fixed the value of the other functions f_k , for $k \neq j$.
- This algorithm produces the same fit of least squares when f_j are regression splines, but the idea is appealing because it can be used with any **generic smoothers** $\mathcal{S}_j$.
- Finally, note that under the constraint $\sum_{i=1}^n f_j(x_{ij}) = 0$ the least square estimate for the **intercept** term is $\hat{\beta}_0 = \bar{y}$, i.e. the arithmetic mean.

The backfitting algorithm (regression)

The backfitting algorithm for additive regression models

1. Initialize $\hat{\beta}_0 = \bar{y}$ and set $f_j(x_j) = 0$, for $j = 1, \dots, p$.
2. Cycle $j = 1, \dots, p$, $j = 1, \dots, p$, ..., until **convergence**:
 - i. Update the k th function by smoothing via $\mathcal{S}_j$ the **partial residuals**, so that

$$\hat{f}_j(x) \leftarrow \mathcal{S}_j \left[\left\{ x_{ij}, y_i - \hat{\beta}_0 - \sum_{k \neq j} \hat{f}_k(x_{ik}) \right\}_{i=1}^n \right].$$

- ii. Center the function by subtracting its mean

$$\hat{f}_j(x) \leftarrow \hat{f}_j(x) - \frac{1}{n} \sum_{i=1}^n \hat{f}_j(x_{ij}).$$

Backfitting: comments and considerations

- The backfitting algorithm, when f_j are modeled as **regression splines**, is known as “Gauss-Seidel”. The **convergence** is **guaranteed** under standard conditions.
- Interestingly, even when $\mathcal{S}_j$ are **smoothing splines** the **convergence** of backfitting is **guaranteed**; the proof for this statement is less straightforward.
- In general, however, there is no theoretical guarantee that the algorithm will ever converge, even though the practical experience suggest that this is **not** a **big concern**.
- When $\mathcal{S}_j$ is a **linear smoother** with smoothing matrix $\mathbf{S}_j$, then by analogy with the previous unit we can define the **effective degrees of freedom** of $\hat{f}_j$ as

$$\text{df}_j = \text{tr}(\mathbf{S}_j).$$

The number of degrees of the whole model therefore is $\text{df} = 1 + \sum_{j=1}^p \text{df}_j$.

- A variant of backfitting for classification problems is available. Once again, relying on **quadratic approximations** of the log-likelihood allows for a generalization to GLMs.

The backfitting algorithm (classification)

Local scoring algorithm for additive logistic regression

1. Initialize $\hat{\beta}_0 = \text{logit}(\bar{y})$ and set $f_j(x_j) = 0$, for $j = 1, \dots, p$.

2. Iterate **until convergence**:

i. Define the quantities $\hat{\eta}_i = \hat{\beta}_0 + \sum_{j=1}^p \hat{f}_j(x_{ij})$ and $\hat{\pi}_i = \{1 + \exp(-\hat{\eta}_i)\}^{-1}$.

ii. Construct the **working response**

$$z_i = \hat{\eta}_i + \frac{y_i - \hat{\pi}_i}{\hat{\pi}_i(1 - \hat{\pi}_i)}, \quad i = 1, \dots, n.$$

iii. Construct the **weights** $w_i = \hat{\pi}_i(1 - \hat{\pi}_i)$, for $i = 1, \dots, n$.

iv. Use a **weighted backfitting** algorithm using the z_i as responses, which produces a new set of estimates $\hat{f}_1, \dots, \hat{f}_p$.

GAM using penalized splines

- A common special instance of GAM occurs when **smoothing splines** are employed. In the regression case, the **backfitting** algorithm implicitly minimizes the following **penalized loss**

$$\mathcal{L}(f_1, \dots, f_p; \lambda) = \sum_{i=1}^n \left\{ y_i - \beta_0 - \sum_{j=1}^p f_j(x_j) \right\}^2 + \sum_{j=1}^p \lambda_j \int_{a_j}^{b_j} \{f_j''(t)\}^2 dt,$$

where $\lambda = (\lambda_1, \dots, \lambda_p)$ is a vector of **smoothing parameters**.

- Each $f_j(x; \beta)$ is a **natural cubic spline**, therefore the penalized least squares criterion is

$$\mathcal{L}(\beta; \lambda) = \sum_{i=1}^n \left\{ y_i - \beta_0 - \sum_{j=1}^p f_j(x_j; \beta_j) \right\}^2 + \sum_{j=1}^p \lambda_j \beta_j^T \mathbf{\Omega}_j \beta_j,$$

whose joint **minimization** over β is available in closed form.

- Hence, a **direct algorithm** that minimizes $\mathcal{L}(\beta; \lambda)$ is used instead of backfitting.

On the choice of smoothing parameters

- In GAMs there are p **smoothing parameters** $\lambda_1, \dots, \lambda_p$ that must be selected. We can proceed in the usual way, e.g. considering the **generalized cross-validation** criteria:

$$\text{GCV}(\lambda_1, \dots, \lambda_p) = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_i}{1 - \text{df}/n} \right)^2.$$

- An alternative criterion in this context is the **REML** (Restricted Maximum Likelihood), which is the **marginal likelihood** of the corresponding **Bayesian model**.
- It is **not possible** to construct a **grid** of values for all the combinations of smoothing parameters $\lambda_1, \dots, \lambda_p$, because the number of terms increases exponentially in p .
- Hence, many software packages **numerically optimize** the $\text{GCV}(\lambda_1, \dots, \lambda_p)$, or other information criteria, as a function of $\lambda_1, \dots, \lambda_p$, using e.g. the Newton-Raphson method.
- Such an approach is particularly convenient in combination with **smoothing splines**, because the **derivatives** needed for Newton's method are available in closed form.

GAM and variable selection

- When p is large there is a need to remove the potentially **irrelevant variables**. There exist several **variable selection** ideas for GAMs, but we will not cover the details here.
- **Option 1. Stepwise regression**. Perhaps the simplest method, although it is not as efficient as in linear models because we cannot exploit the same computational tricks.
- **Option 2. COSSO: Component Selection and Smoothing Operator** (Lin and Zhang, 2006). It's an idea based on combining lasso-type penalties and GAMs.
- **Option 3. SpAM: Sparse Additive Models** (Ravikumar, Liu, Lafferty and Wasserman, 2009). Similar to the above, but it exploits a variation of the non-negative garrote.
- **Option 4. Double-penalty and shrinkage** (Marra and Wood, 2011). It acts on the penalty term of smoothing splines so that high-values of $\lambda_1, \dots, \lambda_p$ leads to constant functions.
- **Option X. Fancy name**. Yet another method for variable selection with GAMs.

GAM modeling of trawl data

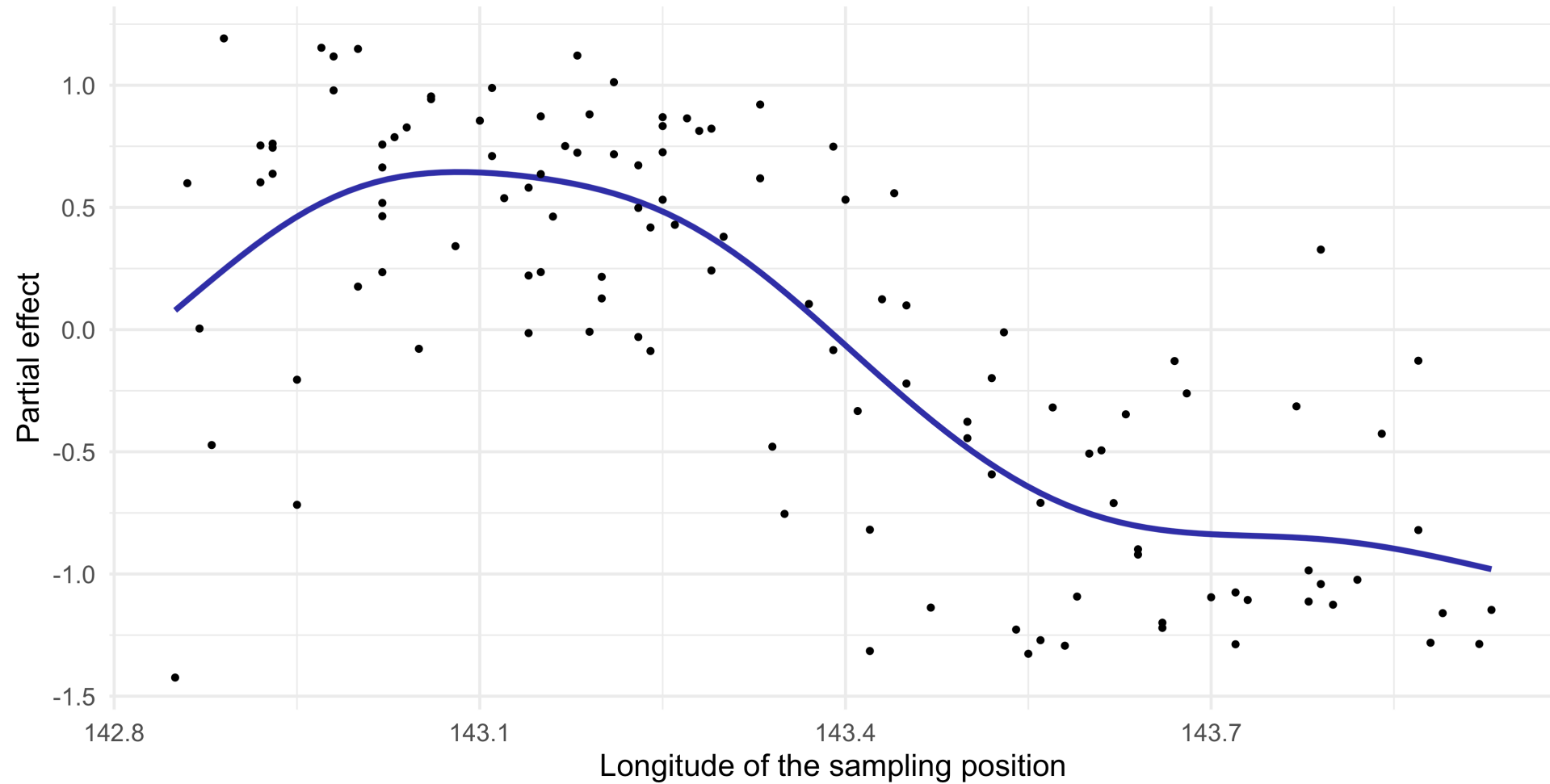
- Let us get back to the trawl data. A specification based on GAM could be

$$\text{Score}_i = \beta_0 + f_1(\text{Longitude}_i) + f_2(\text{Latitude}_i) + f_3(\text{Depth}_i) + \beta_1 I(\text{Zone}_i = \text{Closed}) + \beta_2 I(\text{Year}_i = 1993).$$

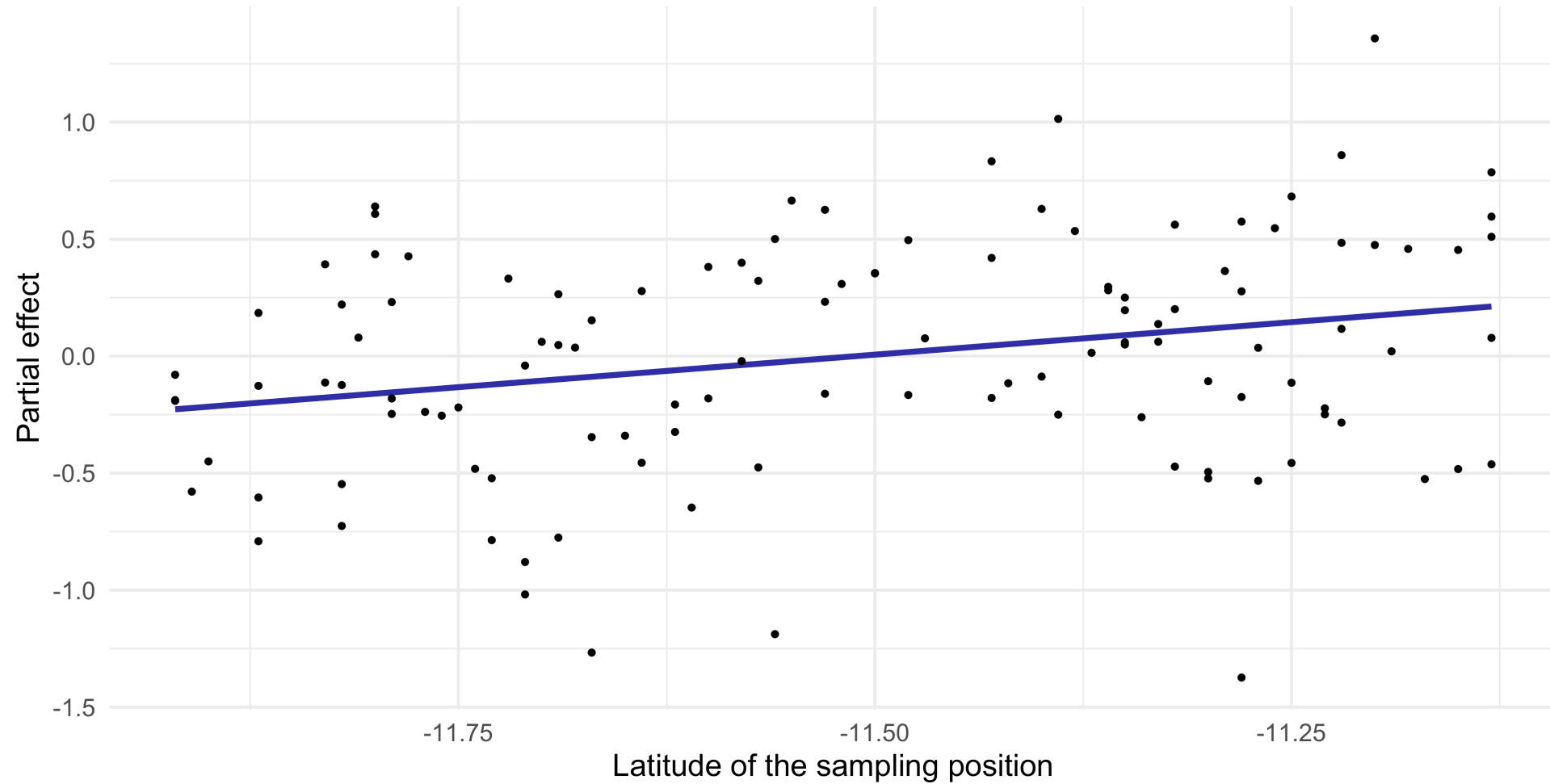
- In GAMs the predictors are **not necessarily** modeled using **nonparametric** methods. Indeed, it is common to have a combination of smooth functions and linear terms.
- Besides, it does **not make sense** to “smooth” a **dummy variable**.

term	estimate	std.error	df
(Intercept)	0.849	0.088	1
Zone_Closed	-0.075	0.099	1
Year_1993	0.149	0.093	1
s(Longitude)	-	-	4.694
s(Latitude)	-	-	1
s(Depth)	-	-	2.447

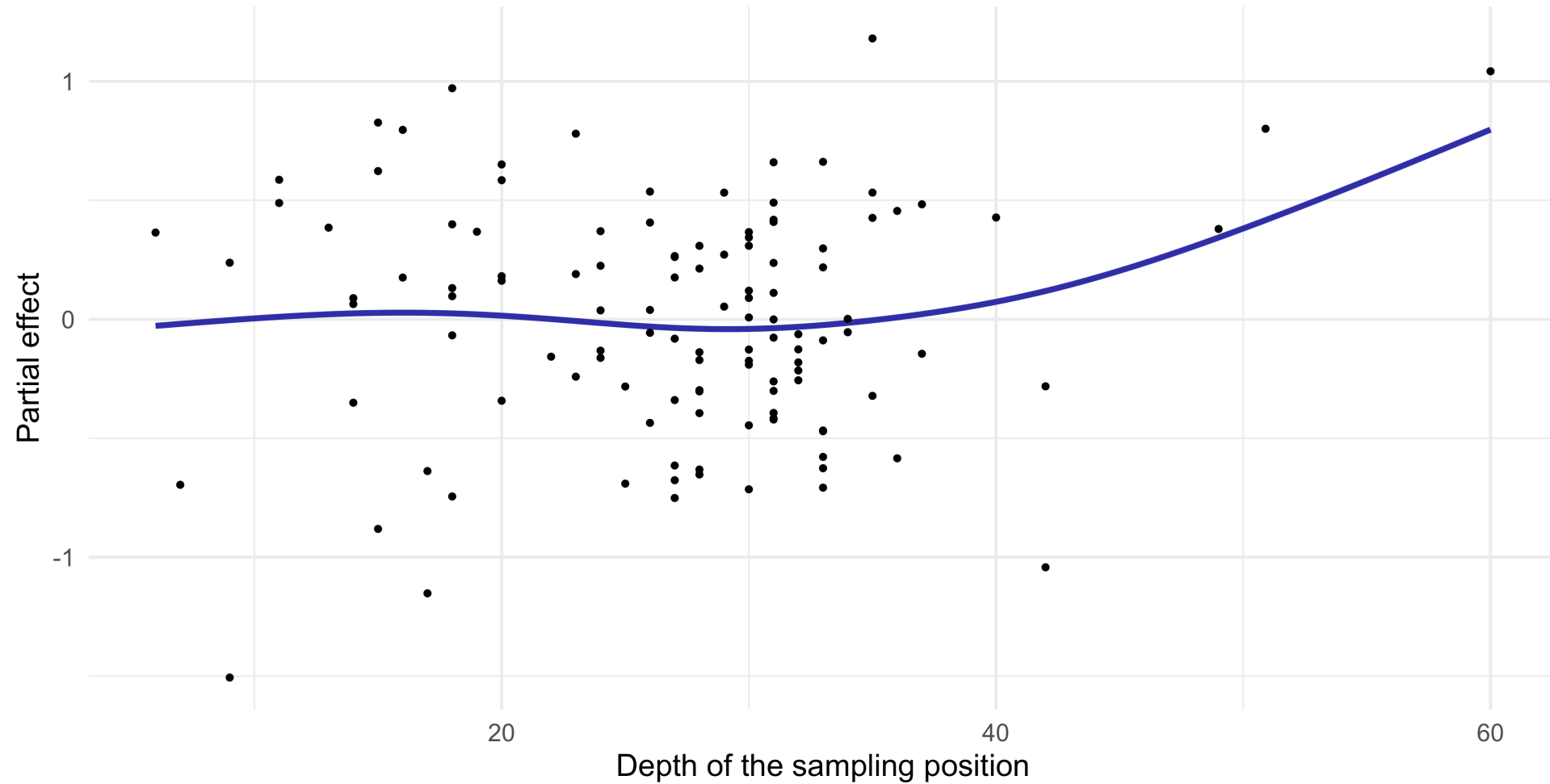
Partial effect of GAMs (Longitude)



Partial effect of GAMs (Latitude)



Partial effect of GAMs (**Depth**)



Comments and criticism (**trawl** data)

- The fitted GAM model highlights some interesting aspects of the **trawl** data.
- In the first place, it seems confirmed that the **Longitude** has a **marked non-linear** impact on the **catch score**, as the initial analysis was suggesting.
- In particular, the catch score is high when the sampling location is close to the coast (but not too close!), and then it suddenly decreases.
- The effective degrees of freedom of **Latitude** is $df_2 = 1$, meaning that the estimated $\hat{f}_2$ **collapsed** to a **linear term**. The corresponding shrinkage parameter λ_2 is very high.
- Overall, the effect due to the **Latitude** looks **small** or **not present** at all.
- The **Depth** seems to have a **relevant effect** on the **Score**, but this is likely due to a few **leverage points** at the right extreme of the **Depth** range.
- Finally, we note that both **Zone** and **Year** seem to have a minor effect.



- Naïve Bayes classifier and GAMs

- The **naïve Bayes classifier** expresses the **binary** classification probability $\text{pr}(y = 1 \mid \mathbf{x})$ as

$$\text{pr}(y = 1 \mid \mathbf{x}) = \frac{\pi_1 \prod_{j=1}^p p_{j1}(x_j)}{\pi_0 \prod_{j=1}^p p_{j0}(x_j) + \pi_1 \prod_{j=1}^p p_{j1}(x_j)} = \frac{\pi_1 \prod_{j=1}^p p_{j1}(x_j)}{p(\mathbf{x})}.$$

- Hence, using class 0 as a **baseline**, we can derive the following expression:

$$\log \frac{\text{pr}(y = 1 \mid \mathbf{x})}{\text{pr}(y = 0 \mid \mathbf{x})} = \log \frac{\pi_1 \prod_{j=1}^p p_{j1}(x_j)}{\pi_0 \prod_{j=1}^p p_{j0}(x_j)} = \log \frac{\pi_1}{\pi_0} + \sum_{j=1}^p \log \frac{p_{j1}(x_j)}{p_{j0}(x_j)} = \beta_0 + \sum_{j=1}^p f_j(x_j).$$

- Therefore, although naïve Bayes and GAMs are fitted in a quite different way, there is a **tight connection** among the two methods.
- Naïve Bayes has a **generalized additive model structure**. This also suggests that the “**additive assumption**” is linked to the notion of **independence** among the covariates.

☠️ - The **mgcv** R package



- GAMs were **invented** by Hastie and Tibshirani in 1986, including the backfitting algorithm.
- Simon Wood (2003) described **thin-plate regression splines** and their estimation (no backfitting).
- Simon Wood (2004, 2011) invented methods for estimating $\lambda_1, \dots, \lambda_p$ in an **efficient** and **stable** manner.
- Marra and Wood (2011) discussed many methods for practical **variable selection** for GAMs.
- For further details, there is a **recent and advanced book** by Simon Wood (2017) entitled “*Generalized Additive Models: An Introduction with R*”.
- The **mgcv** package in **R** (by Simon Wood) implements everything mentioned here.

Pros and cons of generalized additive models (GAMs)

Pros

- GAMs can automatically model **non-linear** relationships. This can potentially make more accurate predictions for the response.
- GAMs, as linear models, are **interpretable**: the variation of the fitted response, holding all but one predictor fixed, **does not depend** on the values of the **other predictors**.
- In practice, this means that we can **plot** the **fitted functions** $\hat{f}_j$ separately to examine the roles of the predictors in modelling the response.
- Additive assumption is quite strong, but it is still possible to **manually add interactions** as in the linear regression case.

Cons

- Especially when p is large, it is almost impossible to manually model all the **interactions** among covariates. GAMs do **not** take **second-order** effects (or higher) into account.

Results on the test set

- We can now check the predictive performance on the **test set**, to see which model scored the best in terms of MAE and RMSE.

	Null model	Linear model	GAM
MAE	0.611	0.361	0.315
RMSE	0.718	0.463	0.408

- It is quite clear that including a **non-linear** specification for the **Longitude** was indeed a good idea that improved the predictive performance.

References

References I

- **Main references**

- **Chapters 4 and 5** of Azzalini, A. and Scarpa, B. (2011), *Data Analysis and Data Mining*, Oxford University Press.
- **Chapter 9** of Hastie, T., Tibshirani, R. and Friedman, J. (2009), *The Elements of Statistical Learning*, Second Edition, Springer.

- **Historical references**

- Hastie, T., and Tibshirani, R. (1986). *Generalized Additive Models*. *Statistical Science* **1**(3), 209-318.

Specialized references I

■ Simon Wood's contributions

- Wood, S. N. (2003). Thin plate regression splines. *Journal of the Royal Statistical Society. Series B: Statistical Methodology* **65**, 95-114.
- Wood, S. N. (2004). Stable and efficient multiple smoothing parameter estimation for generalized additive models. *Journal of the American Statistical Association* **99**, 673-86.
- Wood, S. N. (2011). Fast stable restricted maximum likelihood and marginal likelihood estimation of semiparametric generalized linear models. *Journal of the Royal Statistical Society. Series B: Statistical Methodology* **73**(1), 3-36.
- Marra, G. and Wood, S. N. (2011). Practical variable selection for generalized additive models. *Computational Statistics and Data Analysis* **55**(7), 2372-87.
- Wood, S. N. (2017). *Generalized Additive Models: An Introduction with R*. CRC Press.